RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Monday, 5 October 2026, 9:25 AM
State	Finished
Completed on	Monday, 5 October 2026, 9:32 AM
Time taken	7 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>

int main(){
    int n;
    if(scanf("%d", &n)!=1) return 0;

    int *a= (int *)malloc(n*sizeof(int));
    for(int i=0;i<n;i++){
        if(scanf("%d", &a[i])!=1){
            free(a);
            return 0;
        }
    }

    int k;
    if(scanf("%d", &k)!=1){
        free(a);
        return 0;
    }

    int i=0, j=1, found=0;

    while(i<n && j<n){
        if(i!=j && a[j]-a[i]==k){
            found=1;
            break;
        }
        else if(a[j]-a[i]<k){
            j++;
        }
        else{
            i++;
        }
        if(i==j){
            j++;
        }
    }

```

```
40      printf("%d", found);
41
42      free(a);
43      return 0;
44  }
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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